

FROM DATA TO DECISIONS: PREDICTING SUGARCANE YIELD FOR SMARTER FARMING

Group 09

Presented By

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What Do Farmers Want to Know?

Group 09

 *“Is my sugarcane yield high or low this year?”*

 *“What factors affect my sugarcane yield the most?”*

 *“Can I improve my yield with better decisions?”*

 *“What type of fertilizer works best in my region?”*



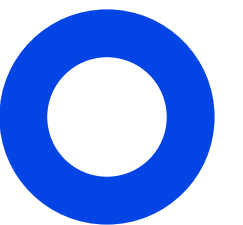
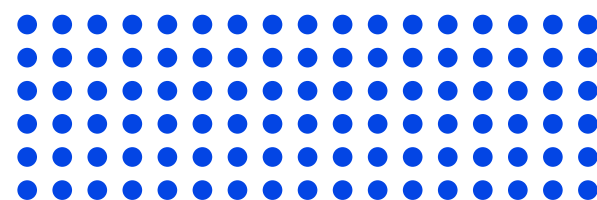
“A farmer’s best yield doesn’t just come from the soil — it comes from the decisions they make.”



Smarter Farming Starts Here: How Our Tool Helps Farmers



- ✓ Predicts whether a farmer's yield is High or Low
- 🌱 Identifies important factors that affect yield.
- 📊 Gives insights based on past data from 8 countries over 25 years
- 💡 Recommends the best fertilizer type based on input
- 🌐 Accessible via website → easy to use for planning and decision-making



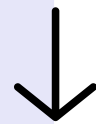
Data set overview

18944 Observations

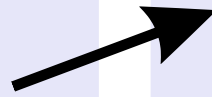
Response variable



Yield



Yield Category



Low

High

Predictor variables



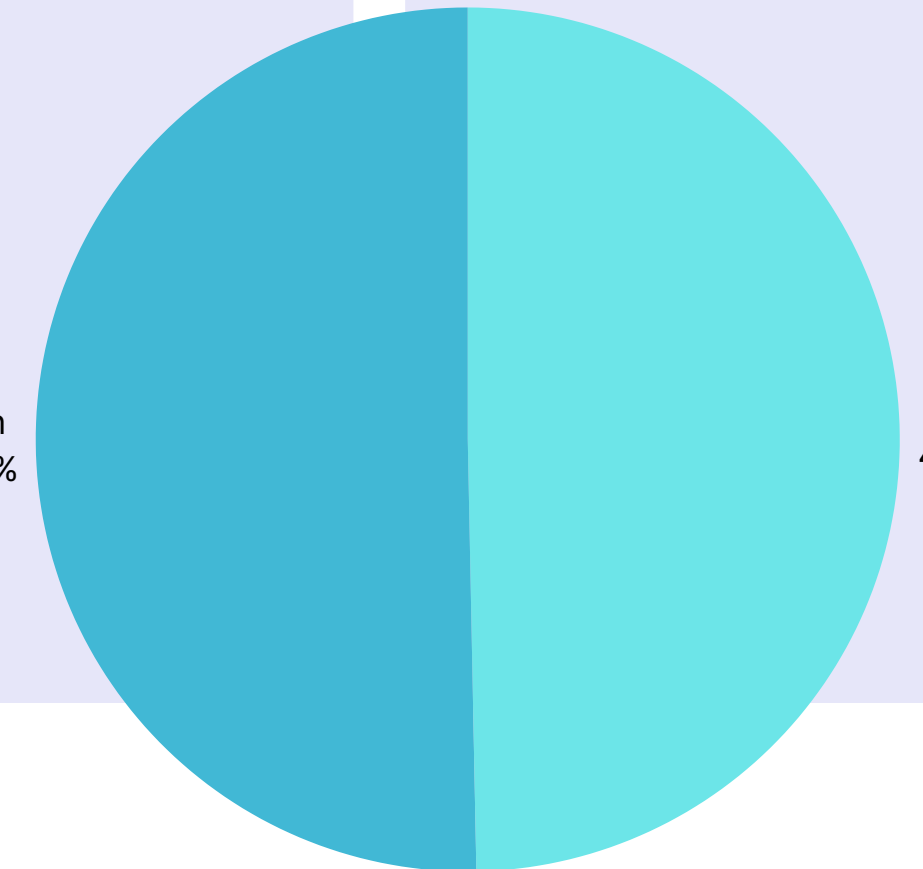
Quantitative : 6



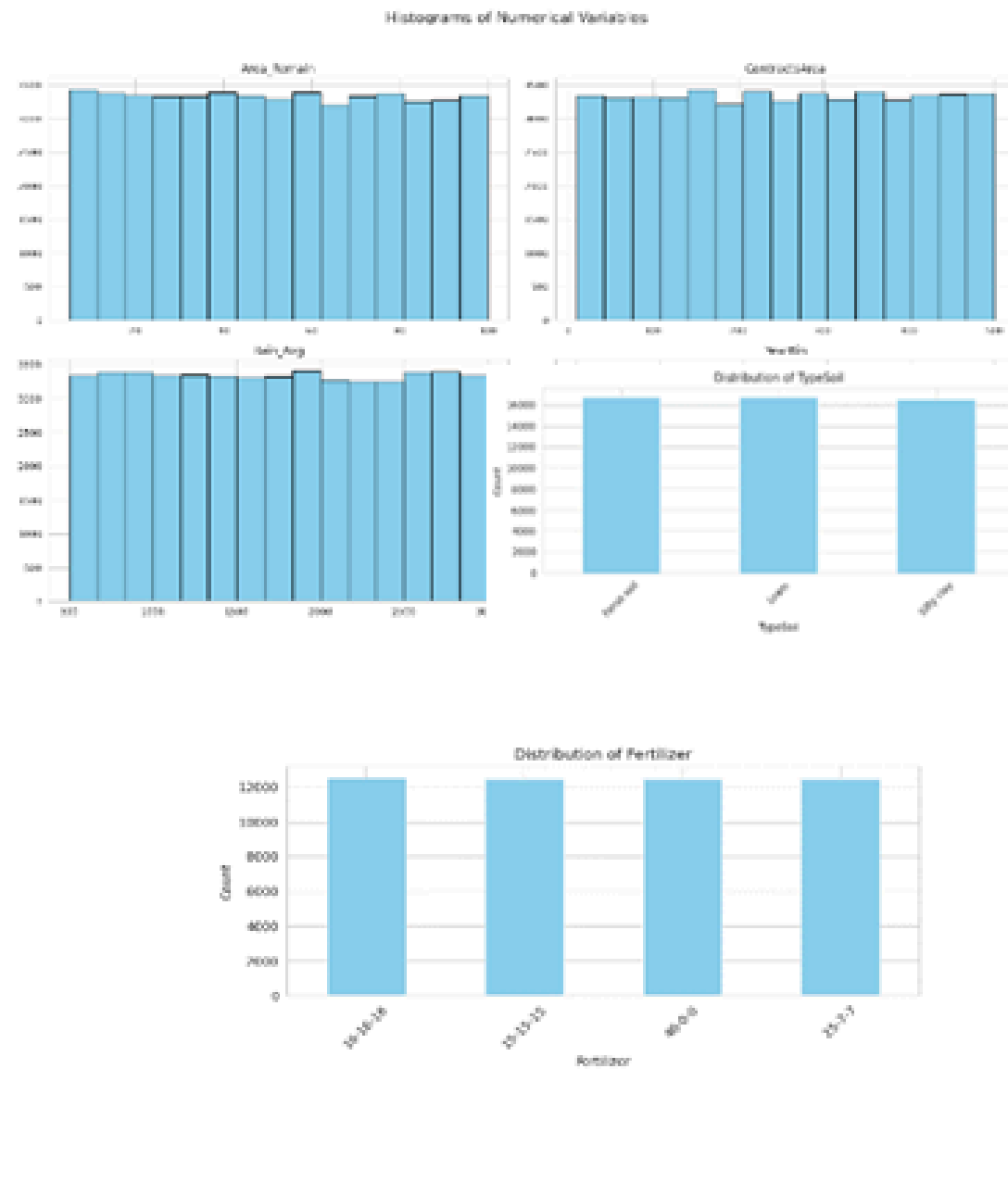
Qualitative : 9

High
50.3%

Low
49.7%



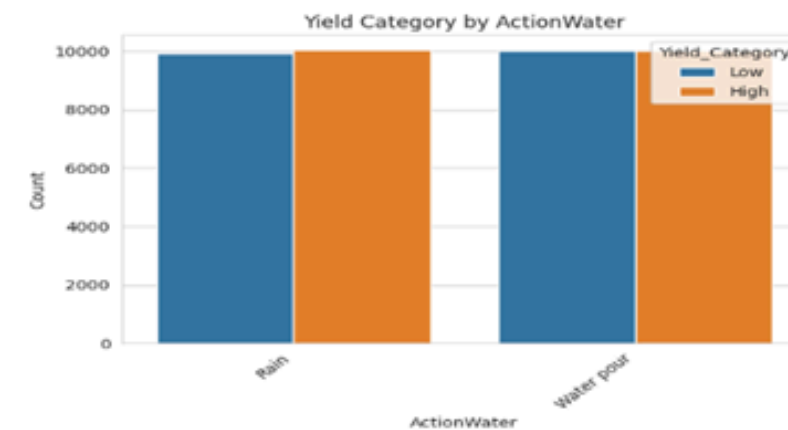
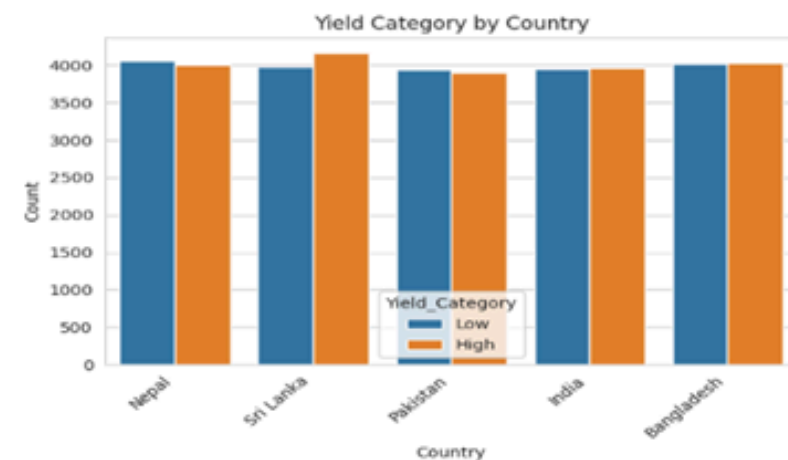
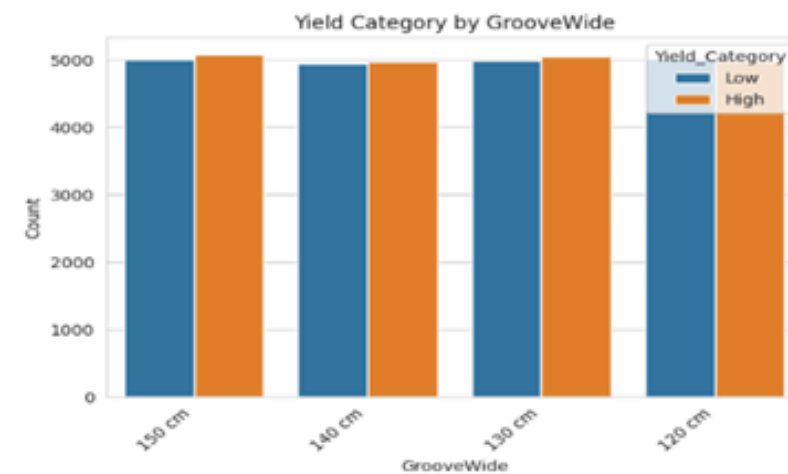
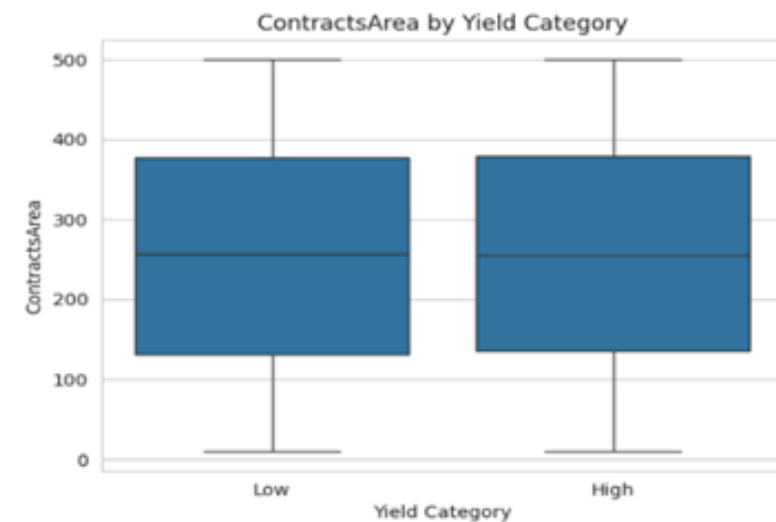
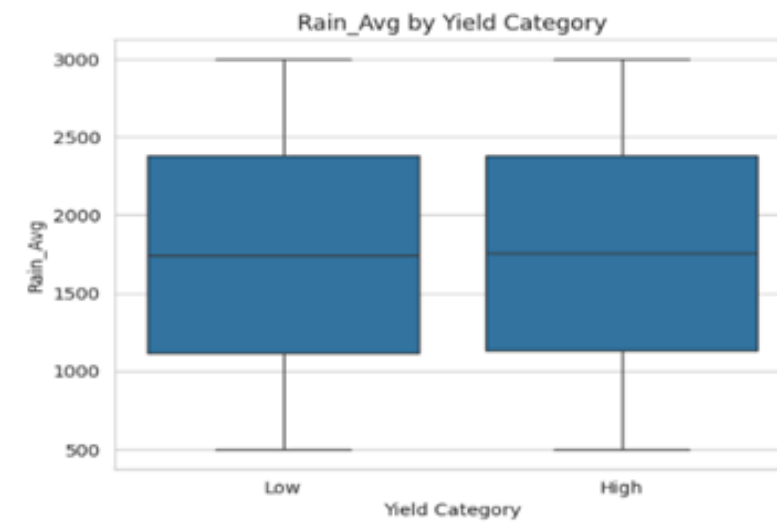
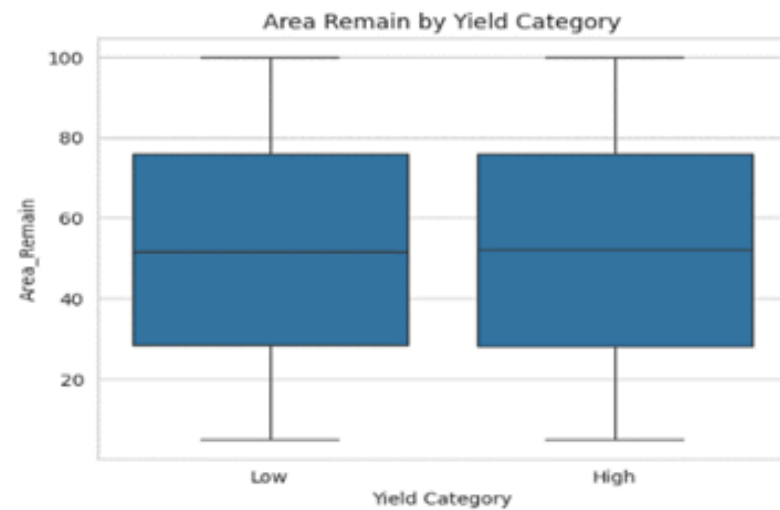
EDA Results and Findings



- There is little or no variance between the categories for each variable

It might not be helpful in predicting the target variable

EDA Results and Findings

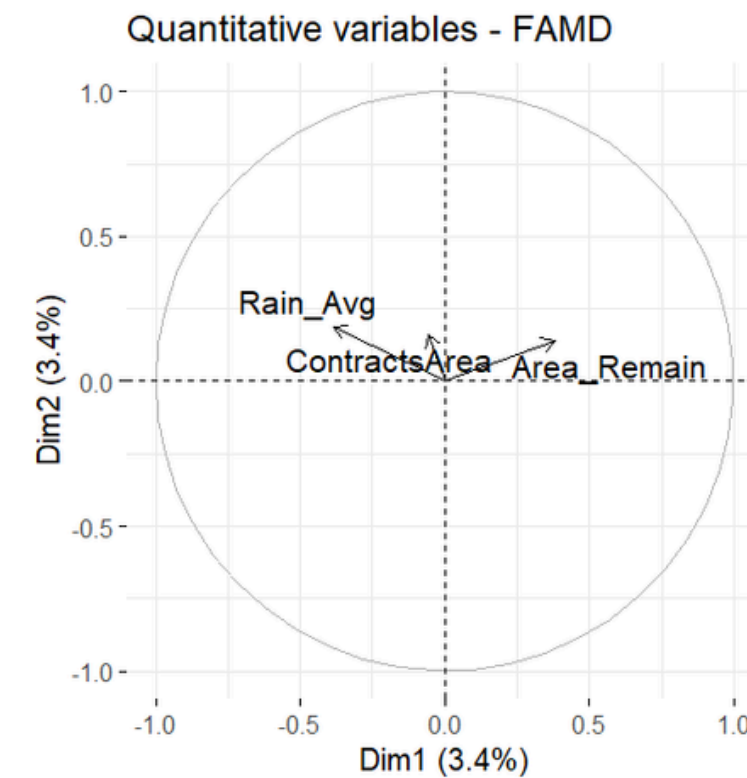
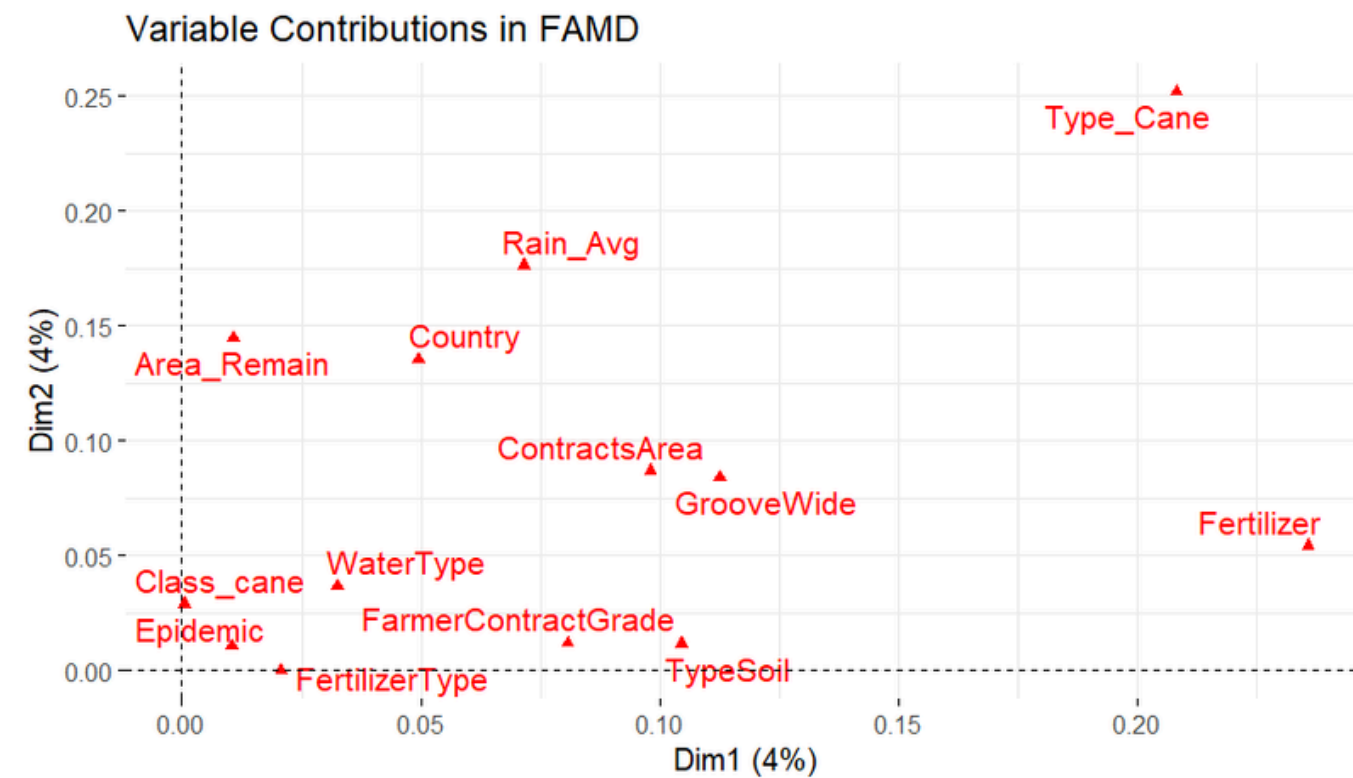


Based on the association between each variable and the Yield categories.

- The median and IQR are same for all numerical variables.
- The distribution of "Low" and "High" yield categories is almost uniform across all the categorical variables.

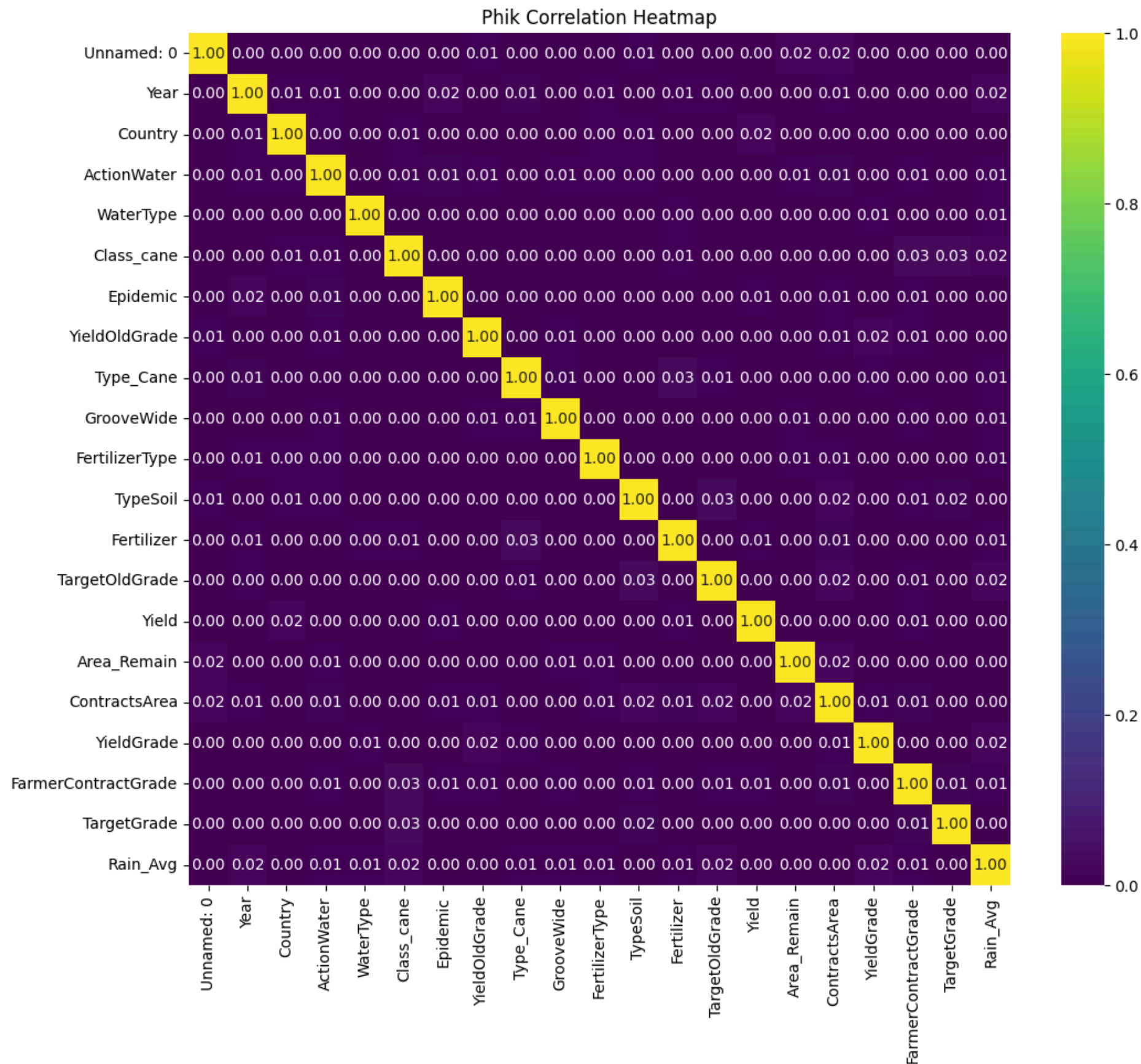
suggest that all most all the variables may not strongly influence the yield category

Searching for Multicollinearity



There is no any strong multicollinearity

SEARCHING FOR MULTICOLLINEARITY



Phi K handles both categorical and continuous variables — making it suitable for our mixed-type dataset.



Most variables show very low correlation (close to 0).



No strong correlation pairs

➤ Multicollinearity is not a concern in this dataset.

➤ We can safely include all variables for model training without collinearity-based variable removal.

HANDLING OUTLIERS IN THE DATASET

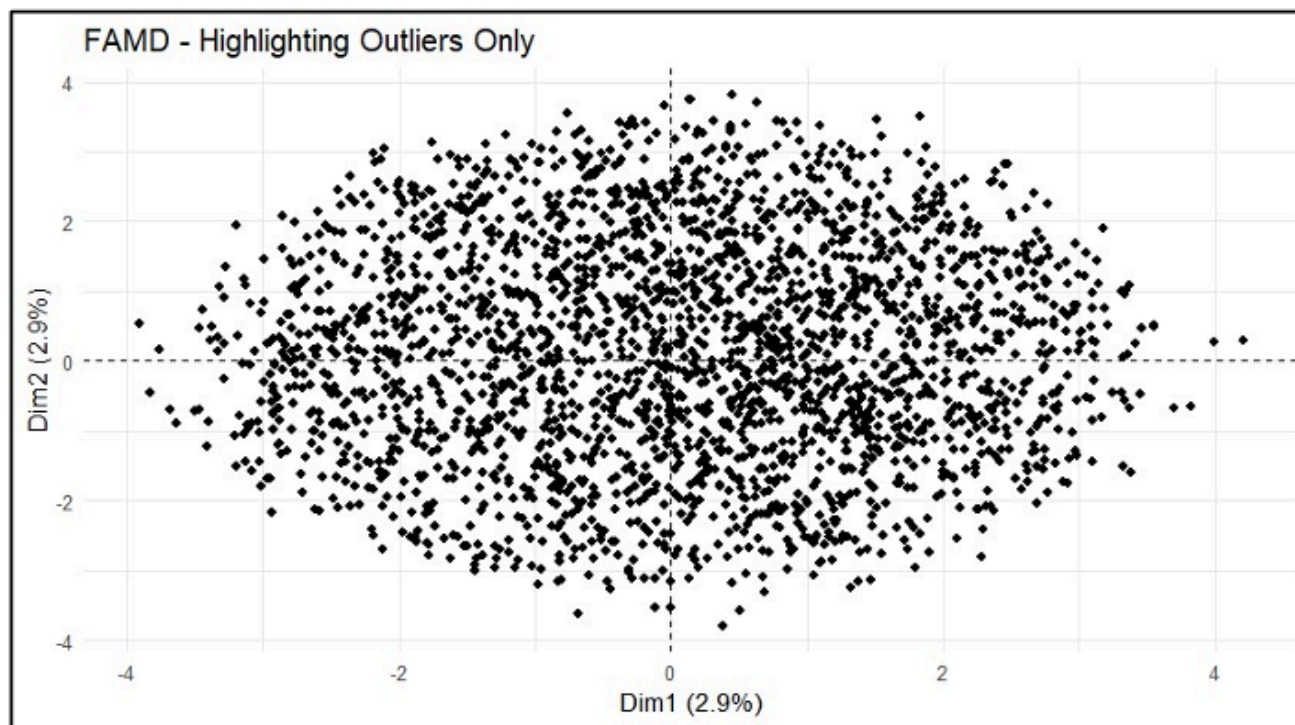
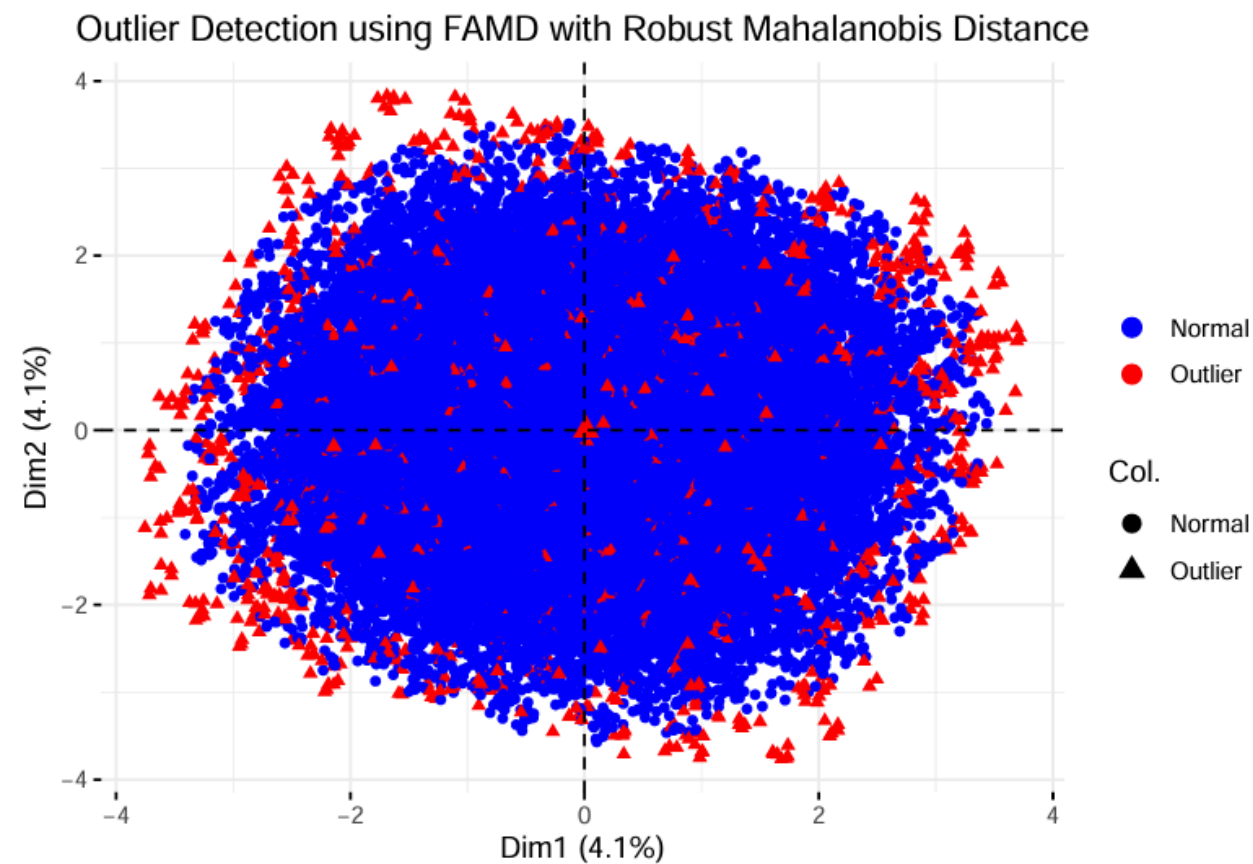
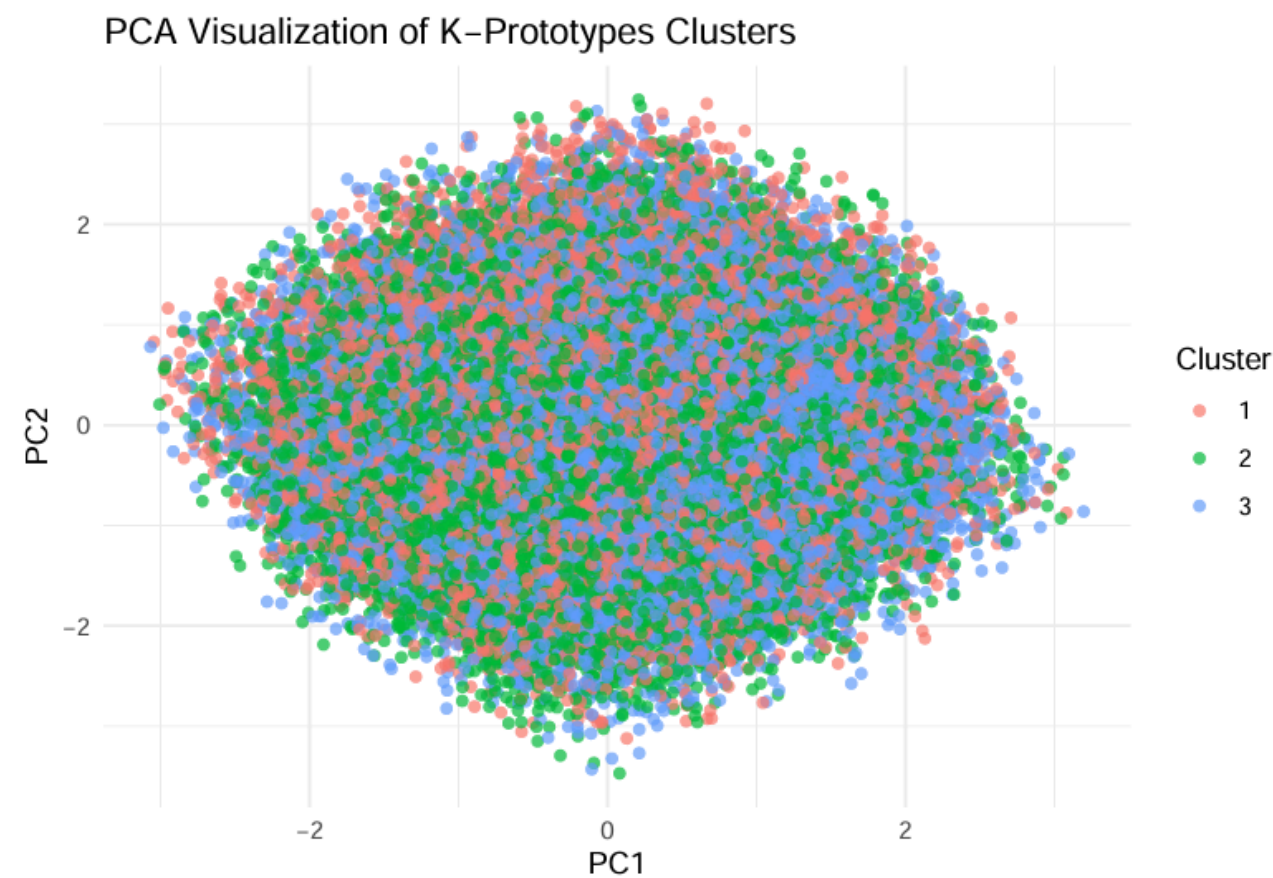
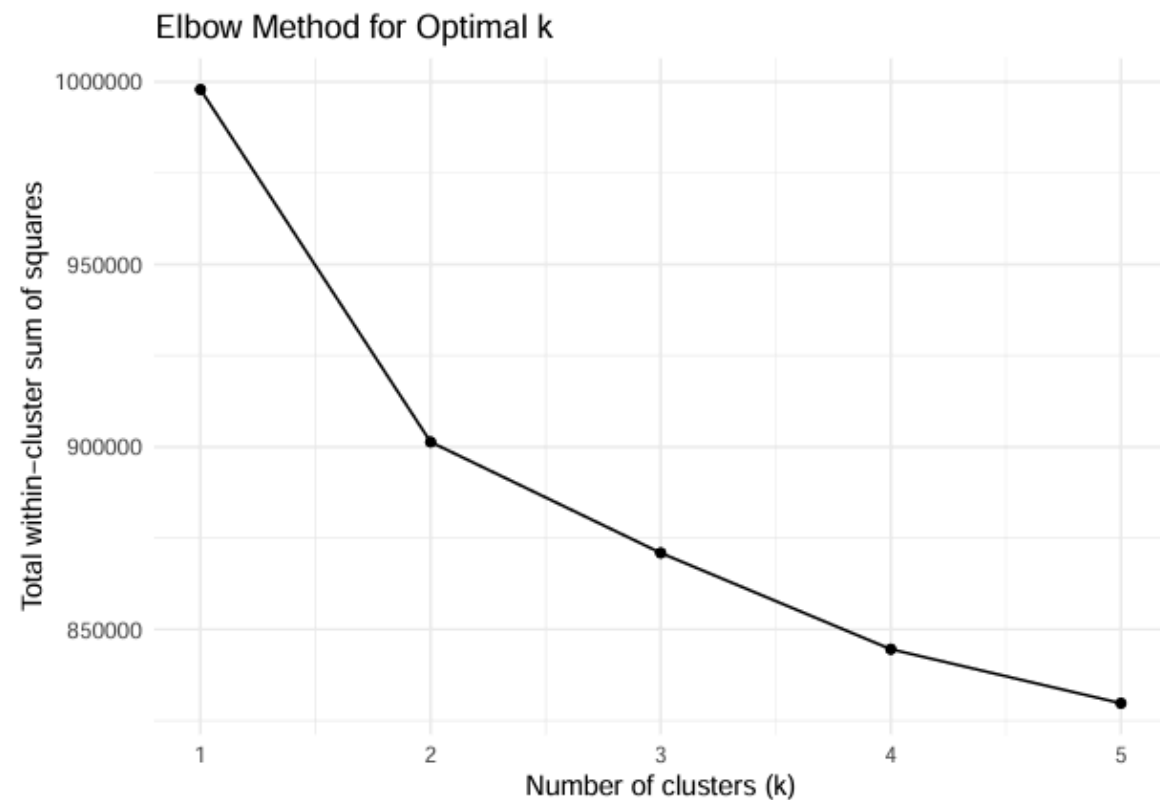


Figure 18 – FAMD-Highlight Outliers Only

- Used **Robust Mahalanobis Distance** and **Isolation Forest**
- Found around **2,000 outliers**.
- Did not remove them because:
 - No strong reason to reject them.
 - Model performance did not improve after removal.
- ✅ Decided to keep outliers in the final dataset

CLUSTERING EXPLORATION AND FINDINGS

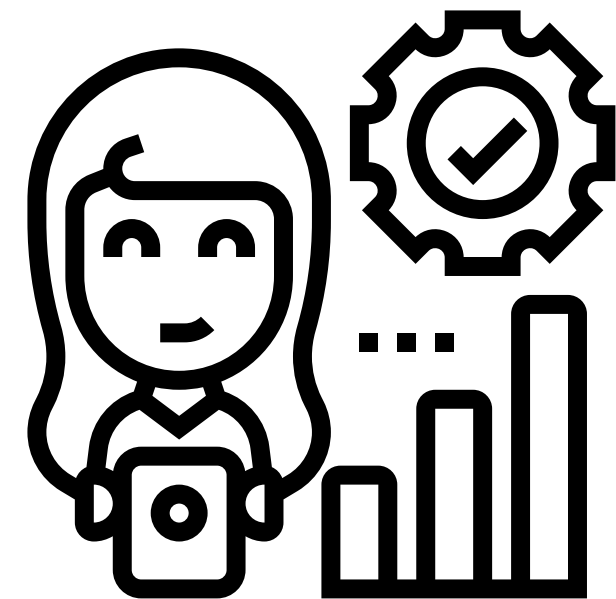


▲ Optimal number of cluster is 2 by elbow method.

▲ By K-Prototypes algorithm Silhouette Score is 0.002.

▲ Clusters are highly overlapping with no clear separation.

✓ There is no strong clustering pattern in the dataset.



MODELING APPROACH

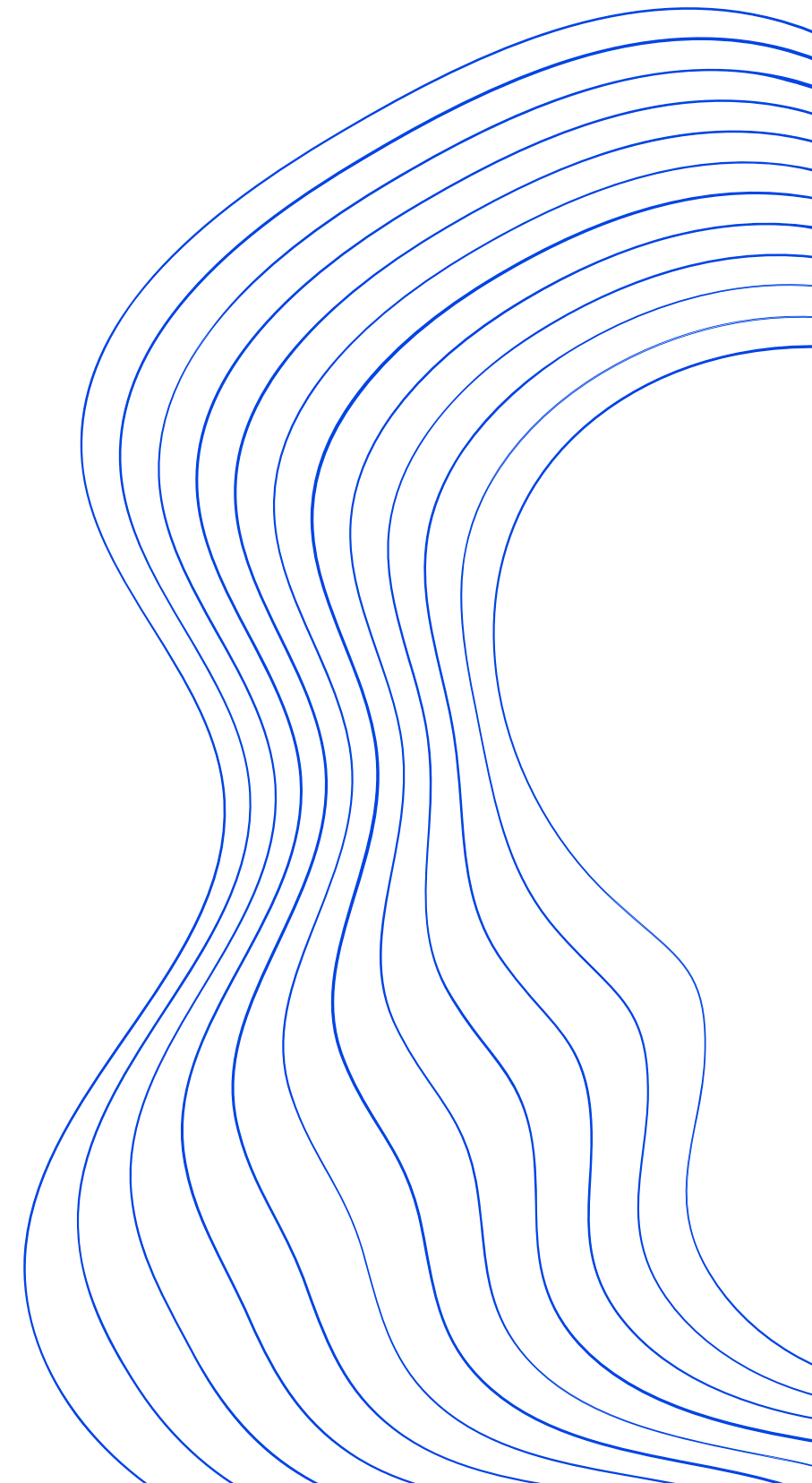


Logistic Regression

Random Forest

Xg Boost

SVM



WHY THESE MODELS

Logistic Regression: More interpretable.

Random Forest: Handles complex interactions, and less sensitive to overfitting.

XGBoost: Handle complex interactions and give high accuracy for classification problems.

SVM: Handle imbalance data and can handle complex data.



MAXIMIZING PERFORMANCE: MODEL PARAMETER TUNING

Model	Tuning Method	Best Parameters
Random Forest	Grid Search	ntree = 500, mtry = 9
XGBoost	Grid Search	eta = 0.1, max_depth = 6, nrounds = 500
Logistic Regression	Grid Search	$\alpha = 0.5$, $\lambda = 0.01$
SVM	Manual / Auto selection	kernel = linear

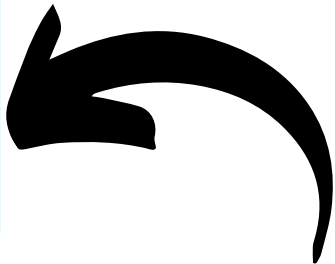


“Each model was carefully tuned to ensure balanced accuracy and optimal performance.”



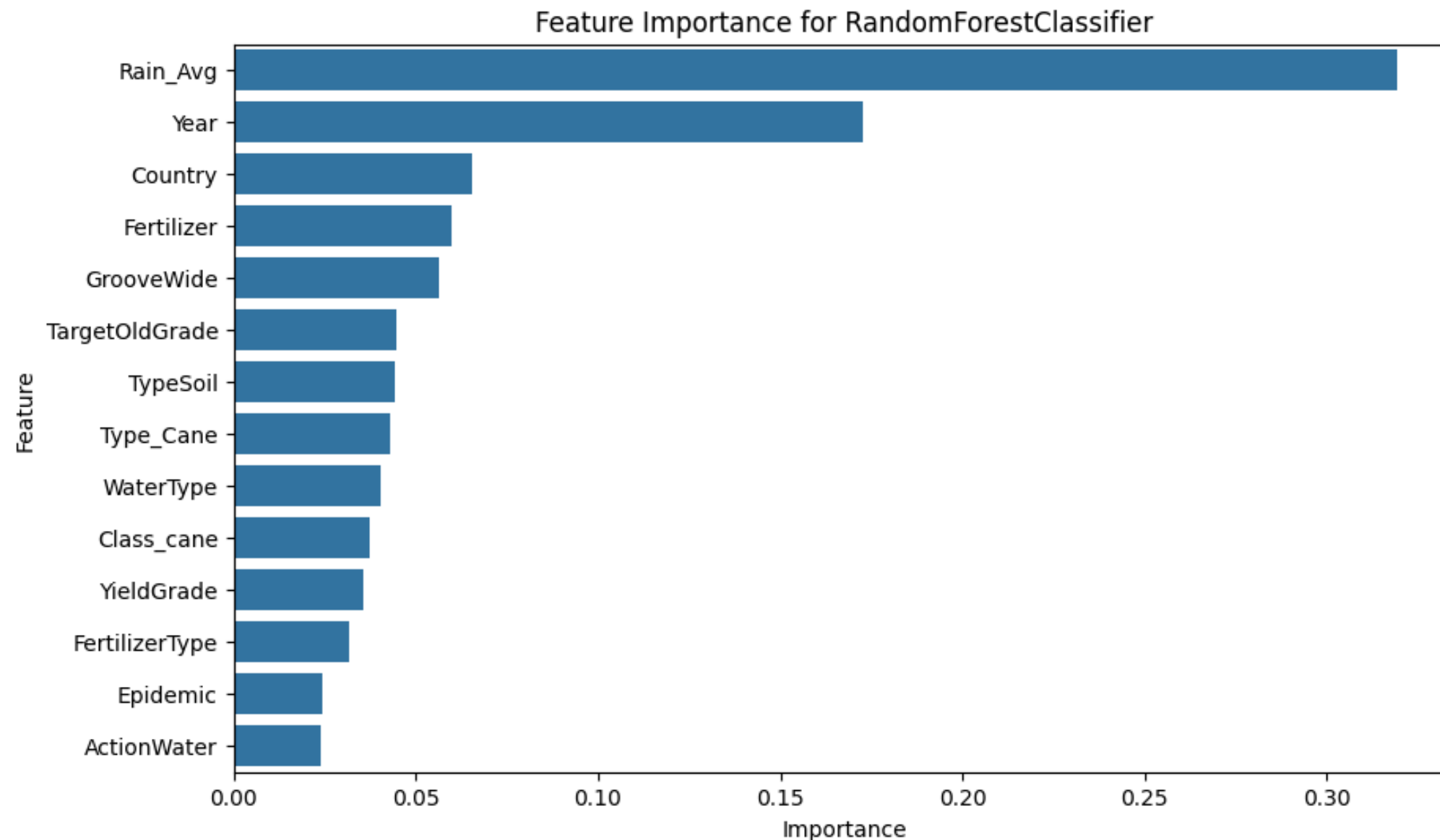
MODEL PERFORMANCE COMPARISON

Model	Accuracy (Train)	Accuracy (Test)	F1 Score (Test - Low)	F1 Score (Test - High)
Logistic	0.5052	0.5049	0.4700	0.5400
Random Forest	0.5292	0.5077	0.5100	0.5000
XG Boost	0.5755	0.5107	0.4586	0.5536
SVM	0.5117	0.4985	0.4923	0.5046



Best model

WHAT DRIVES OUR PREDICTIONS ?



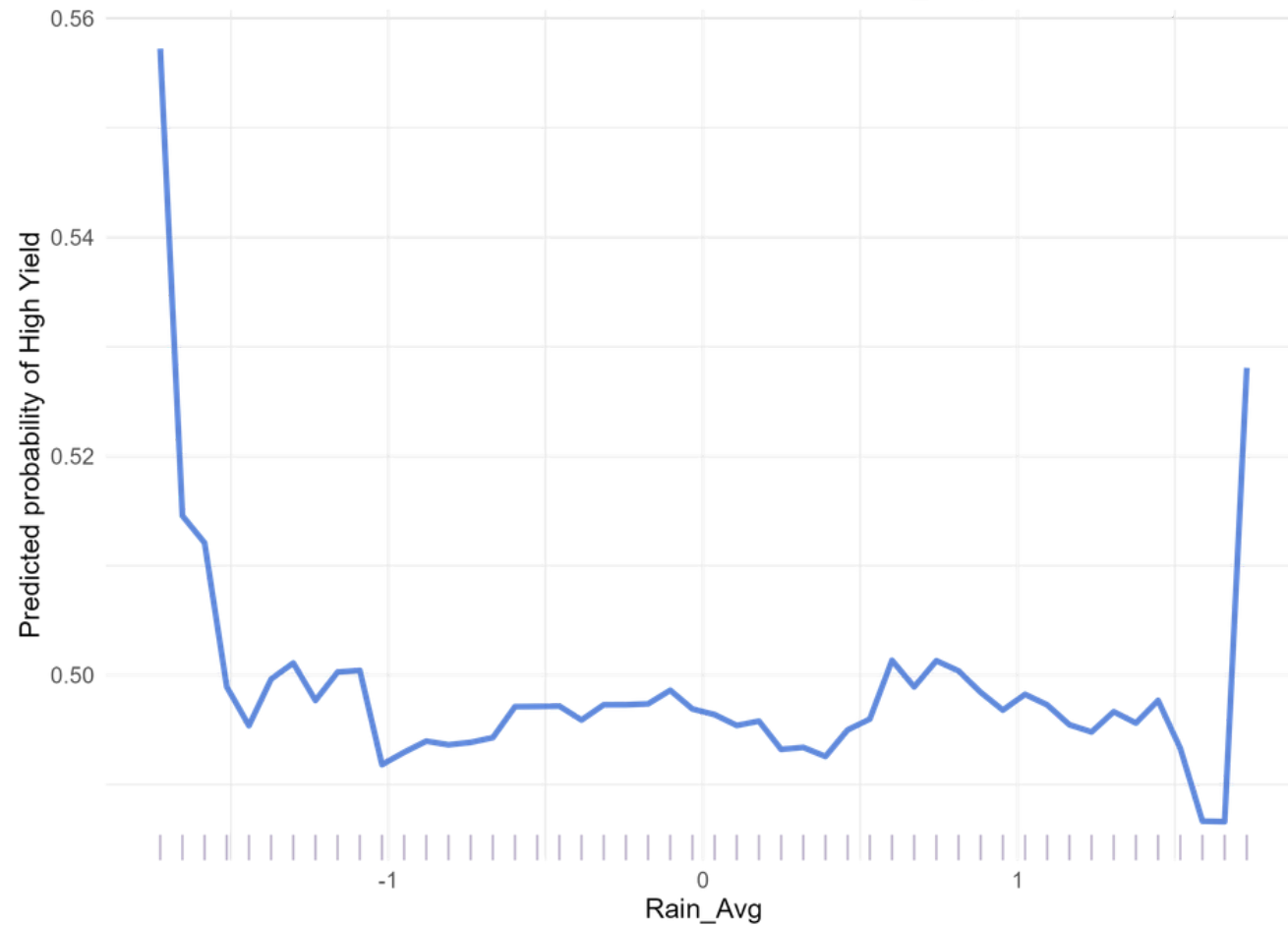
- **Top predictors:** Rain_Avg and Year are by far the most important features.
- **Secondary factors:** Country, Fertilizer, and GrooveWide have noticeable but smaller effects.
- **Moderate influence:** Variables like TargetOldGrade, TypeSoil, Type_Cane, and WaterType have moderate-to-low impact.

Insight: Environmental conditions are the key drivers of yield variation.

HOW FEATURES INFLUENCE SUGARCANE YIELD PREDICTIONS



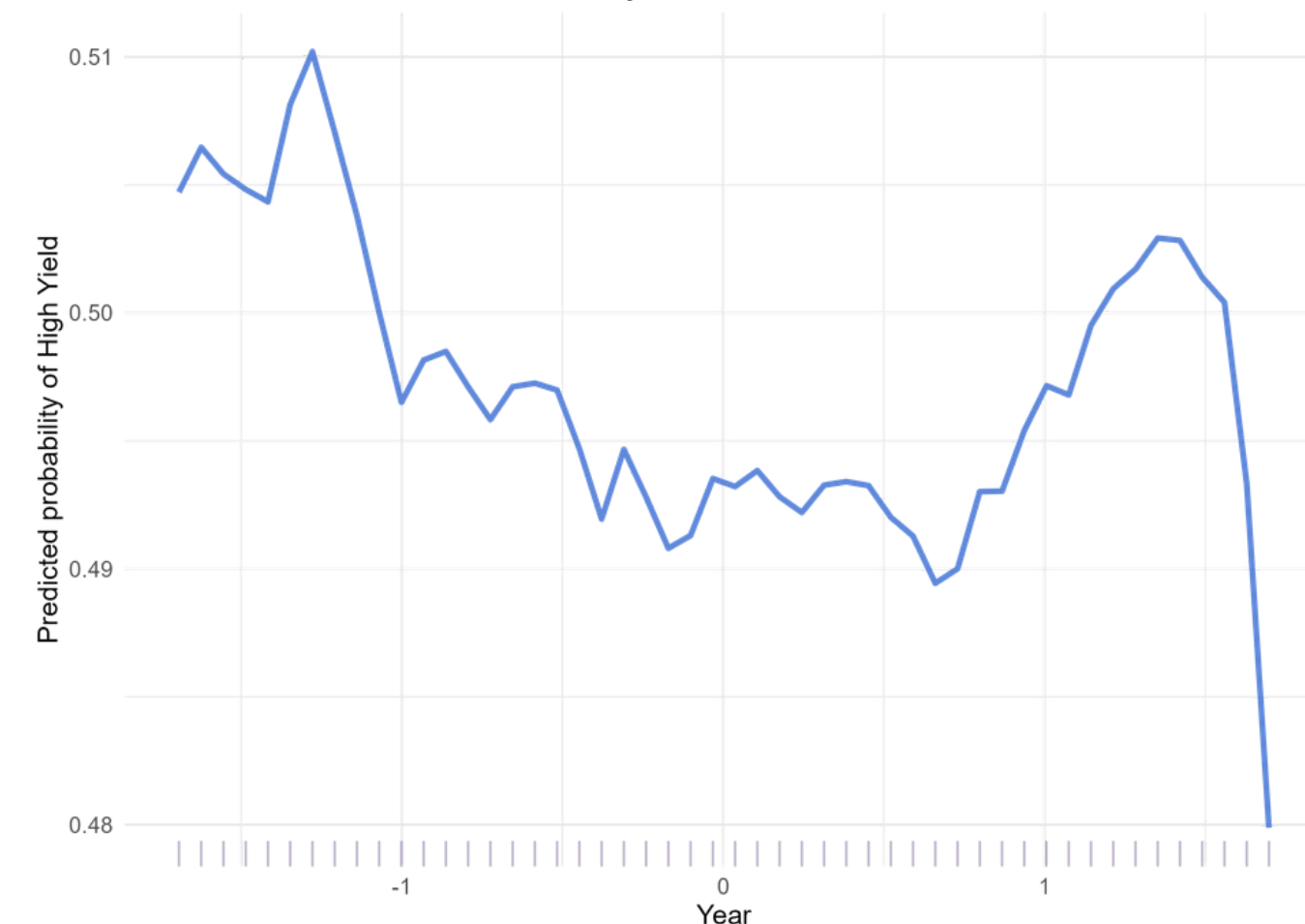
Partial Dependence Plot for Rain_Avg



- Very low and Very high rainfall predicts highest yields.
- Mid-range rainfall shows consistently lower yield probability.

- No clear trend in the plot.
- Middle period shows lowest yield probability (around 0.49)

Partial Dependence Plot for Year



FACTOR ANALYSIS FOR MIXED DATA

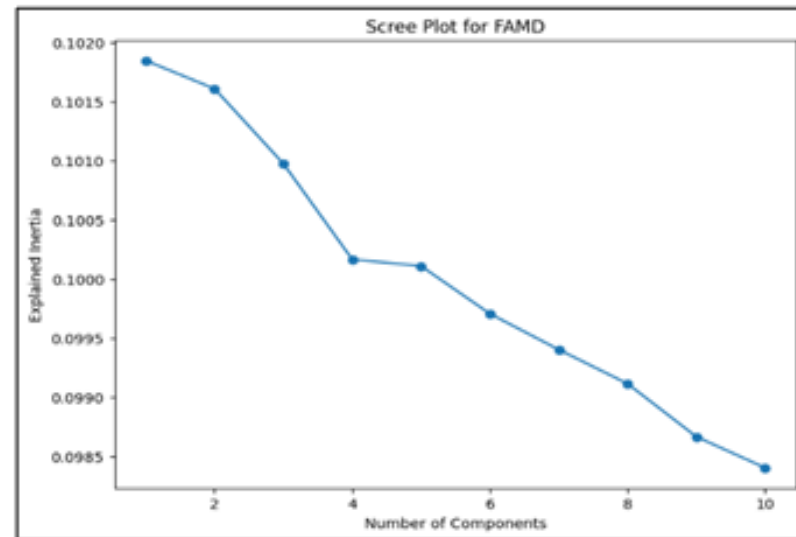


Figure 19 – Scree plot for FAMD

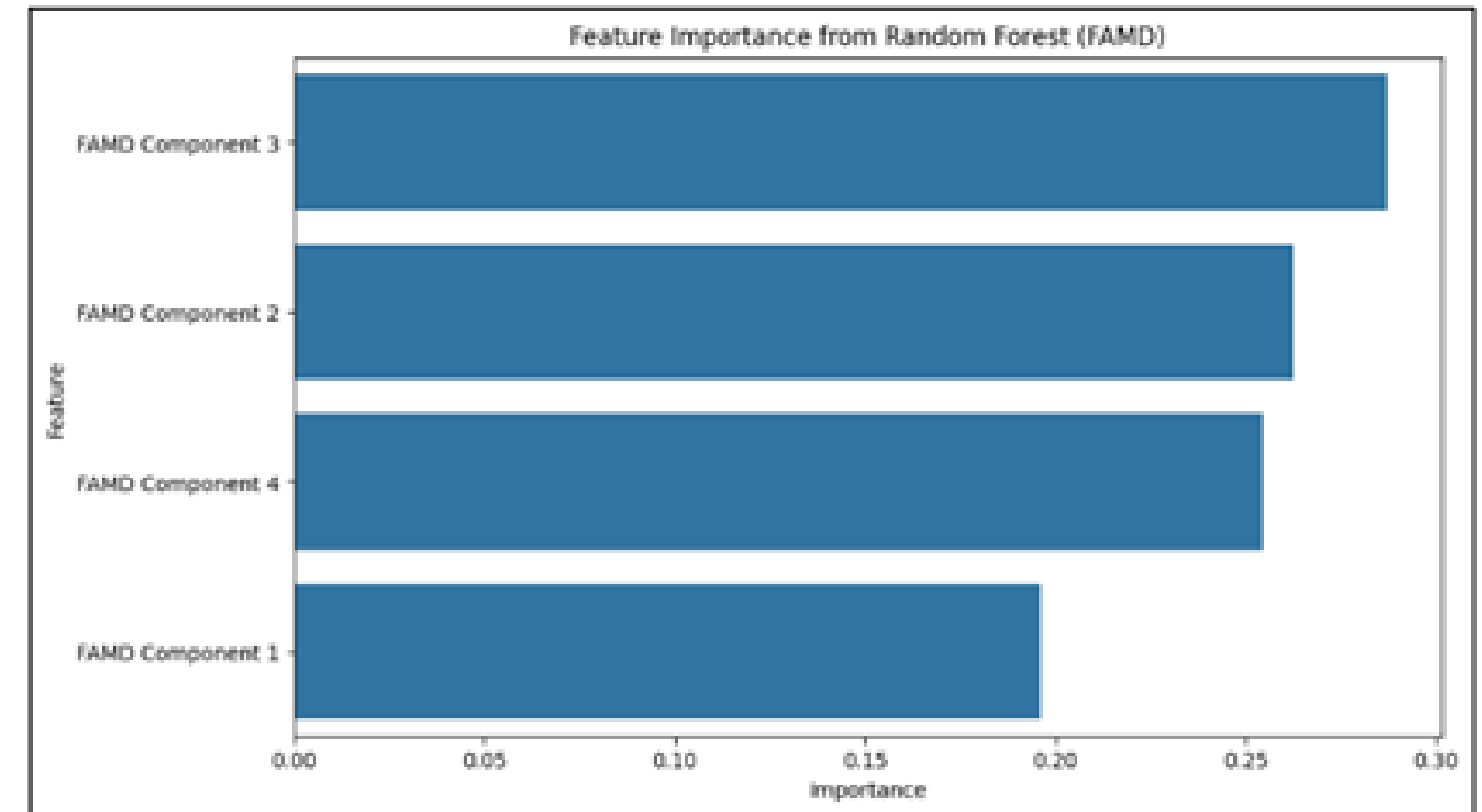
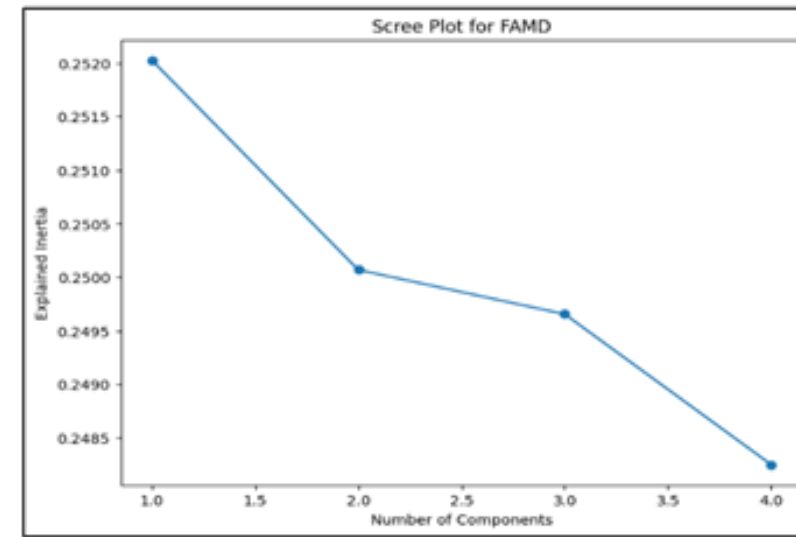


Figure 21 – Feature Importance for Random Forest

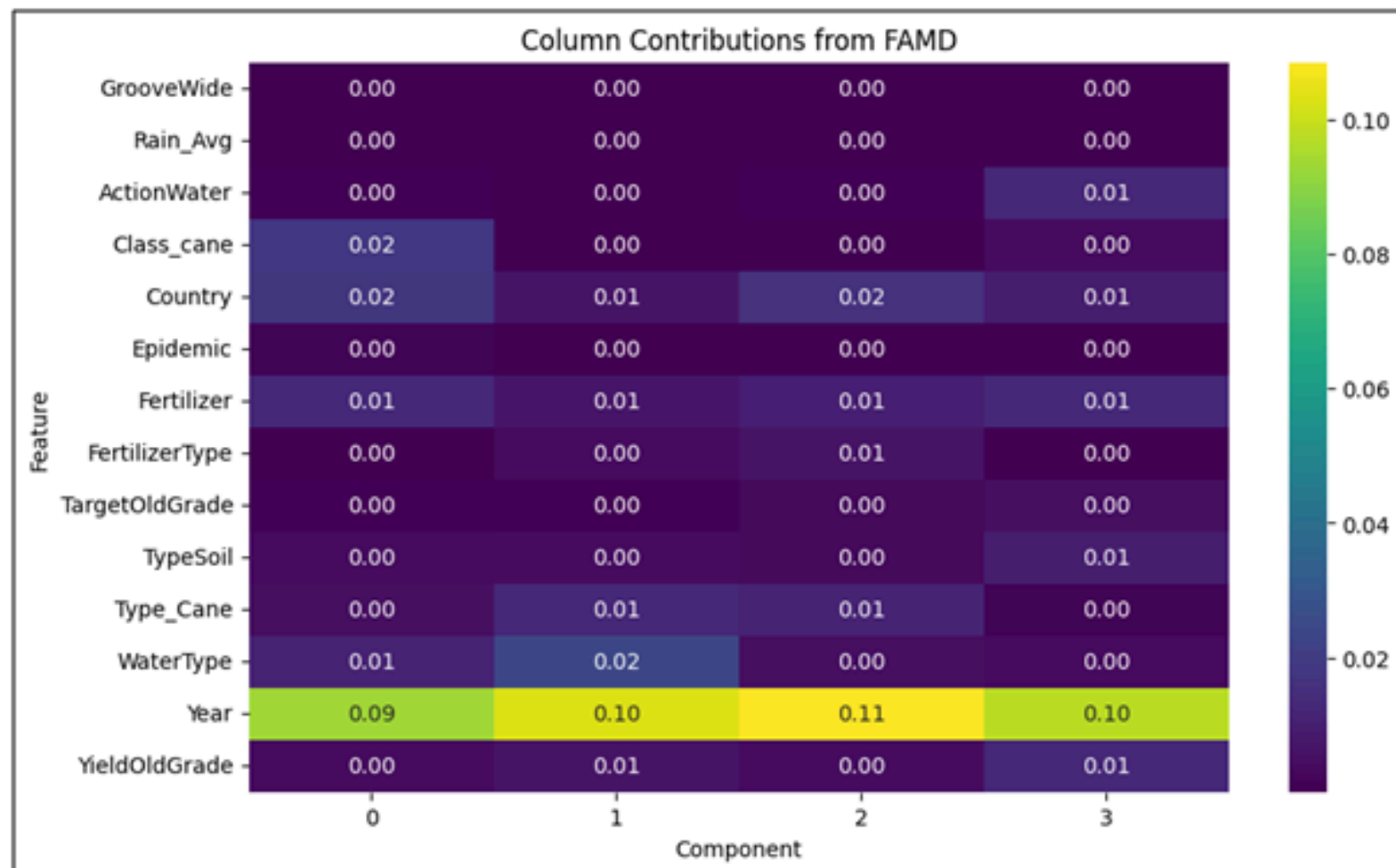


Figure 20 – Column Contribution for FAMD

Component 1: Year to year trends connected with cane harvest ratoon and country.

Component 2: Year to year irrigation type differences.

Component 3: Regional treatment interaction over time. (Technological improvements included)

Component 4: Climate and weather based seasonal or cyclic trends.

KEY CHALLENGES IN DATASET ANALYSIS



**Low Prediction
Power**



**Irrelevant
Variables**



**Weak
Correlation with
Response**



**Model
Performance
Issues**

FUTURE IMPROVEMENTS & CONCLUSION

Better Data Collection:

Future improvements could include collecting more detailed, real-world farming data with relevant predictive variables.

Enhanced Variables:

Including additional factors like soil quality, weather patterns, and regional agricultural practices could improve prediction accuracy.

Advanced Models:

Exploring more complex models or ensemble methods could further enhance predictive performance.

Long-Term Vision:

With improved data and modeling techniques, this tool could become a valuable resource for farmers, helping them make more informed decisions on sugarcane yield.



Concluding Thoughts

Even though our current model isn't perfect, this project lays the foundation for a tool that can eventually help farmers make better decisions, save resources, and improve yields.



Predict Your Sugarcane Yield with Precision

Harness AI and data analytics to forecast your harvest accurately and optimize your farming practices.

Get Started!



Accurate Predictions



Optimized Farming



Regional Focus

REFERENCES

- freeCodeCamp.org. (2022, October 6). Flask Tutorial - Full Course for Beginners (2022) [Video]. YouTube. https://youtu.be/dam0GPOAvVI?si=g00LjITaVhwd_LQw
- Python Simplified. (2021, March 31). Deploy Flask app to Heroku in 15 minutes — Tutorial [Video]. YouTube. <https://youtu.be/jQjjqEjZK58?si=pXXzcuMcs4iLHm7p>
- freeCodeCamp. (2018, October 22). How to Build a Web Application Using Flask and Deploy it to the Cloud. <https://www.freecodecamp.org/news/how-to-build-a-web-application-using-flask-and-deploy-it-to-the-cloud-3551c985e492/>
- https://www.youtube.com/channel/UCoCToADdJRd3u-ACz4e_iCw
- Thambavita, P., & Samarasinghe, G. (2024). A novel method to predict sugarcane yield using hybrid machine learning approach. BIO Web of Conferences, 67, 00157. <https://doi.org/10.1051/bioconf/20246700157>
- Sugarcane Research Institute. (n.d.). Sugarcane Research Institute Sri Lanka. <https://sugarres.lk/>
- Chelaru, N. (n.d.). FAMD: Factor Analysis of Mixed Data. RPubS. <https://rpubs.com/nchelaru/famd>
- Nair, M. (2020, August 12). Feature selection: Mutual Information. Medium. <https://medium.com/@miramnair/feature-selection-mutual-information-a0def943e1ed>

THANK YOU

FOR YOUR ATTENTION

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